Course: Introduction to Artificial Intelligence
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MapOut – AI-Powered Travel Planner

Week 5: Informed Search and Heuristics

This week's planned objective

Study informed search algorithms and heuristics, expand the MapOut attraction dataset to multiple destinations across North Macedonia, and implement a simple baseline for generating travel itineraries based on user preferences, budget, and available time.

What I completed

I studied informed search, heuristic functions, Greedy Best-First Search, and A*, including examples of how these algorithms could be applied to MapOut.

I expanded the dataset across eight destinations and implemented data loading and validation. I also developed a rule-based baseline that generates itineraries based on destination, interests, budget, and available visit time. I tested the baseline using three scenarios.

Evidence submitted

- Week 5 report with theoretical explanations and implementation details.
- Expanded **attractions.csv** dataset with 101 attractions.
- Python code for data loading, validation, and itinerary generation.
- Three test scenarios with screenshots of the generated itineraries.
- GitHub repository containing the Week 5 implementation.

What I can demonstrate now

I can demonstrate how MapOut loads and validates the dataset and generates basic itineraries using predefined user preferences. The results show the selected attractions, total estimated cost, and total visit duration while respecting the specified constraints.

Problem or failure encountered

Some attractions have unknown estimated costs, so the baseline skips them instead of assuming they are free. The current baseline also does not calculate travel time between attractions or optimize their visiting order.

Decision made this week

I decided to expand MapOut beyond Ohrid to include additional destinations across North Macedonia. I also decided to support multiple categories per attraction and begin with a simple itinerary-generation baseline before implementing more advanced search algorithms.

AI assistance used

I used ChatGPT (OpenAI) to understand informed search algorithms, organize the report, and assist with dataset expansion, Python implementation, and testing.

Feedback received

To be completed after receiving feedback.

Next week's objective

Implement the first project-specific AI method using constraint satisfaction and planning. I plan to explore a backtracking approach that selects suitable attractions while respecting the user's interests, budget, and available time.